Monu Singh

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Research Interests

Relativistic hot accretion flows around black holes, dissipative mechanisms in accretion flows, thermal conduction effects, mass outflows from accretion disks, numerical modeling of black hole physics

Education

- **Ph.D. in Astrophysics** July 2018 – November 2025
  Indian Institute of Technology Guwahati, India
- Thesis: "Relativistic Accretion Flows Around Rotating Black Holes: Effects of Viscosity, Thermal Conduction, and Magnetic Fields"
- Advisor: Prof. Santabrata Das
- Focus: Numerical modeling of dissipative processes around black hole accretion with relativistic effects
- **M.Sc. in Physics** 2016
  University of Delhi, India
- **B.Sc. (Hons) Physics** 2013
  University of Delhi, India

Research Experience

Doctoral Research 2018 – 2025
Indian Institute of Technology Guwahati, India *Advisor: Prof. Santabrata Das*

- Developed numerical models for relativistic hot accretion flows around rotating black holes
- Investigated role of thermal conduction in magnetized accretion flows with shock waves
- Studied dissipation mechanisms and viscosity effects in black hole accretion systems
- Analyzed mass outflow properties from magnetized accretion disks

Publications

1. **Singh, M.**, & Das, S. (2024). "Properties of relativistic hot accretion flow around rotating black hole with radially varying viscosity", *Astrophysics and Space Science.* DOI – 10.1007/s10509-023-04263-6 | arXiv:2312.16001
2. **Singh, M.**, & Das, S. (2025). "Role of thermal conduction in relativistic hot accretion flow around rotating black hole with shock", *Journal of Cosmology and Astroparticle Physics (JCAP).* DOI – 10.1088/1475-7516/2025/02/068 | arXiv:2408.02256
3. **Singh, M.**, Jana, C., & Das, S. (2025). "Effect of thermal conduction on accretion shocks in relativistic magnetized flows around rotating black holes", *Journal of Cosmology and Astroparticle Physics (JCAP).* DOI – 10.1088/1475-7516/2025/05/055 | arXiv:2502.16829
4. Jana, C., **Singh, M.**, Rakshit, S., & Das, S. (2025). "Study of mass outflows from magnetized accretion disks around rotating black holes with thermal conduction", *Journal of Cosmology and Astroparticle Physics (JCAP).* DOI – 10.1088/1475-7516/2025/10/090 | arXiv:2503.06209
5. **Singh, M.**, Garain, S., & Das, S. (2026) "Hybrid disc geometry for shocked accretion flows: Unveiling QPOs in black hole X-ray binaries ", *Monthly Notices of the Royal Astronomical Society (MNRAS).* DOI – 10.1093/mnras/stag1574 | arXiv:2608.18615

Conference Presentations

Contributed Talks

- "Properties of relativistic hot accretion flow around rotating black hole with radially varying viscosity," RETCO-V, Indian Institute of Astrophysics, Kodaikanal, India, April 3-5, 2023
- "Dissipative accretion flow around black holes with radially varying viscosity," 32nd Meeting of IAGRG, IISER Kolkata, India, December 19-21, 2022
- "Dissipative accretion flows around black holes with viscosity parameter α as a function of radial distance," NEMA-VIII, Manipur University, India, November 2020

Workshop Participation

- International Workshop on Numerical Astrophysics, Ferdowsi University of Mashhad, Iran, July 2020 (Online)

Poster Presentations

- "Effect of thermal conduction in relativistic accretion flows around black holes," 44th Meeting of Astronomical Society of India, IIT Guwahati, India, May 2026

Awards and Honors

- University Grant Commission National Eligibility Test (NET) & Junior Research Fellowship (JRF) in Physics (2018)

Technical Skills

Programming Languages: Python, Fortran, Mathematica, C
Computational Skills: High-Performance Computing, Numerical Methods
Analysis & Visualization: Data Visualization, Mathematical Modeling
Research Areas: Accretion Physics, General Relativity, Magnetohydrodynamics, Computational Astrophysics

Teaching Experience

Graduate Teaching Assistant 2018 – 2023
Indian Institute of Technology Guwahati *Department of Physics*

- PH101 Tutorials and Peer Assisted Learning (B.Tech): Classical Mechanics, Special Relativity, Quantum Mechanics
- PH102 Tutorials (B.Tech): Electrodynamics
- PH403 Tutorials (M.Sc): Classical Mechanics
- PH413 General Physics Lab (M.Sc): Laboratory instruction and supervision
- PH415 Tutorials (B.Tech): Simulation Techniques in Physical Systems

Professional Service & Leadership

- Conference Host: 44th Meeting of Astronomical Society of India (ASI) 2026, IIT Guwahati, India, May 2026
- Conference Host: 10th International Conference on Gravitation and Cosmology (ICGC) 2023, IIT Guwahati, December 2023
- Conference Host: North East Meet of Astronomers (NEMA-VI), IIT Guwahati, November 2020 (Online)
- Weekly Group Meeting Organizer, IIT Guwahati (2021-2022)